# A Zero-BRAM, Single-Cycle Streaming 2D Convolution Accelerator with Saturating ReLU Datapath

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Ahmed S. Bassala

Department of Electronics and Communications Engineering
Egypt-Japan University of Science and Technology (E-JUST)

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# Architecture: Zero-BRAM & Zero-DSP Strategy

**Distributed Shift-Register Line Buffers (SRL32)**

- $32 \times 32$ input frames with $3 \times 3$ kernel ($N = 3$, $S = 1$).
- Dual line buffers require $2 \times (W - 3) = 58$ words.
- Synthesized to native **SRL32E primitives** (16 LUTs as memory).
- **Eliminates Block RAM (BRAM $= 0$)**, completely avoiding the heavy $100\times$ FoM penalty.

**Pure Fabric Multipliers**

- Enforced via `(* use_dsp = "no" *)`.
- Avoids the $50\times$ DSP denominator penalty ($9 \times 50 = 450$ equivalent LUTs saved).

**Streaming Control Without FSM Overhead**

- Self-synchronizing valid-strobe pipeline.
- Input strobe `s_valid` clocks coordinate counters and shift lines.
- Eliminates multi-state FSM decode logic and branch delay hazards.
- Sustains **1.0 output pixel / cycle** throughput.

**FoM-Driven Architectural Constraint**

Cost $=$ LUTs $+ 50 \times$ DSPs $+ 100 \times$ BRAMs
Zero DSPs and zero BRAMs minimize total cost to raw fabric LUTs (769).

# Datapath Design, Latency Budget & Verification

### Dynamic Range & Bit-Width Analysis

- Pixel input: 8-bit unsigned $[0, 255] \rightarrow$ 9-bit signed positive.
- Weights: 8-bit signed $[-128, +127]$.
- Products: Registered into `logic signed [15:0] prod[3][3]`.
- 4-level balanced adder tree accumulates into an exact **20-bit signed sum** ($[-293760, +291465]$).
- Post-processing: Saturating clamp to $[-32768, +32767]$ cascaded into sign-bit zero-mux ReLU.

### Deterministic Latency Breakdown

- Input fill blanking: $2W + 2 =$ **66** cycles.
- Pipeline registers: **3** cycles (`win_valid`, `prod`, `m_data`).

### Systematic Verification Summary

- Bit-accurate Python NumPy golden reference model.
- Self-checking SystemVerilog testbench (`tb_conv3x3.sv`).
- Evaluated across 7 targeted corner-case suites:

| Test ID | Targeted Behavior | Result |
|---|---|---|
| TC-01 | Dynamic Weight Config Port | **PASS** |
| TC-02 | Spatial Alignment & Row Skew | **PASS** |
| TC-03 | Signed Sobel-X Convolution | **PASS** |
| TC-04 | Signed Sobel-Y Convolution | **PASS** |
| TC-05 | Laplacian Operator & ReLU | **PASS** |
| TC-06 | Max-Stress Range Saturation | **PASS** |
| TC-07 | 900-Pulse Boundary Blanking | **PASS** |

**Scoreboard Match: 900/900 Valid Outputs (0**

# Post-Implementation Physical Characterization

**Artix-7 (`xc7a35tcpg236-1`) Out-of-Context**

| Resource / Metric | Used | Available | Utilization |
|---|---|---|---|
| Slice LUTs | **769** | 20,800 | 3.70% |
| Slice Registers (FFs) | **317** | 41,600 | 0.76% |
| Occupied Slices | **238** | 8,150 | 2.92% |
| LUT as Memory (SRL) | **16** | 9,600 | 0.17% |
| DSP48E1 Blocks | **0** | 90 | **0.00%** |
| Block RAM (RAMB18/36) | **0** | 50 | **0.00%** |
| Worst Negative Slack (WNS) | | **+0.001 ns** | Setup Met |
| Worst Hold Slack (WHS) | | **+0.147 ns** | Hold Met |
| Failing Endpoints | | **0** | Clean Closure |
| Target Clock Period ($T_{clk}$) | | **10.000 ns** | **100.0 MHz** |

**Vivado Power Analysis Breakdown**

- Total On-Chip Power: **0.075 W (75 mW)**
- Dynamic Power: **0.007 W (9%)**
- Device Static Power: **0.068 W (91%)**

## Timing Closure Strategy

- Balanced 4-level adder tree limits combinational logic path.
- Implemented with Vivado physical optimization: `Performance_ExplorePostRoutePhysOpt`.
- Closes timing at 100 MHz with strictly zero negative slack and 147 ps hold margin.

# Competition Summary & Figure of Merit (FoM)

**Table 1: Official Competition Scorecard**

| Parameter | Result | Units | Status / Notes |
|---|---|---|---|
| Input Frame Resolution | $32 \times 32$ | pixels | Fully streaming |
| Input / Weight Precision | 8b / 8b | bits | Unsigned / Signed |
| Architecture Type | Line-Buffer + Win | — | Zero BRAM, pure fabric |
| Multipliers / MACs | 9 parallel | units | Signed LUT multipliers |
| Pipeline Latency | 69 | cycles | Deterministic warm-up |
| Streaming Throughput | 1.0 | pixel/cycle | Sustained single-cycle |
| Hardware Footprint | 769 LUTs, 317 FFs | — | 0 DSPs, 0 BRAMs |
| Clock Frequency | 100.01 | MHz | WNS = +0.001 ns (Met) |
| Total On-Chip Power | 75 | mW | Vectorless report |
| Verification Scoreboard | 100% Match | — | 0 mismatches / 900 px |

**Figure of Merit (FoM) Derivation**

$$\text{FoM} = \frac{\text{Throughput}}{\text{Power} \times (\text{LUTs} + 50\text{DSPs} + 100\text{BRAMs})}$$

$$\text{Cost} = 769 + 50(0) + 100(0) = 769$$

$$\text{Power} = 0.075\,\text{W}$$

$$\text{Throughput} = 1.0\,\text{pixel/cycle}$$

**Official Submission FoM**

$$\text{FoM} = \frac{1.0}{0.075 \times 769} = \mathbf{0.01734}$$

$(1.734 \times 10^{-2}\ \text{pixel}/(\text{W} \cdot \text{Cost}))$